

1.

Full Loss Operator

We know $S_{t+\Delta} = S_t e^{\pi}$

$$\text{Given } V_{t+\Delta} = M[\lambda S_t e^{\pi} - P^{BS}(t+\Delta, S_t e^{\pi})]$$

$$V_t = M[\lambda S_t - P^{BS}(t, S_t)]$$

$$\text{Loss : } L = -(V_{t+\Delta} + V_t)$$

$$L_{t+\Delta}(\pi) = -M[\lambda S_t (e^{\pi} - 1) - (P^{BS}(t+\Delta, S_t e^{\pi}) - P^{BS}(t, S_t))]$$

Linearized Loss Operator

Solve f_t :

$$f(t, z) = M(\lambda e^z - P^{BS}(t, e^z))$$

Since λ is fixed,

$$f_t = -M \frac{\partial P^{BS}}{\partial t}$$

$$\text{Given : } \theta(t, s) = \frac{\partial P^{BS}}{\partial t}$$

$$\text{So, } f_t = -M \theta$$

Solve f_z

$$f_z = M \left[\lambda e^z - \frac{\partial P^{BS}(t, e^z)}{\partial z} \right]$$

$$\frac{\partial P^{BS}}{\partial z} = \frac{\partial P^{BS}}{\partial S} \frac{\partial S}{\partial z}$$

$$\text{Since } \frac{\partial P^{BS}}{\partial S} = \delta$$

$$\text{and } S = e^z \Rightarrow \frac{\partial S}{\partial z} = S$$

$$\text{so } \frac{\partial^2 P}{\partial z^2} = \delta S$$

hence

$$f_z = M[\lambda S - \delta S]$$

$$f_z = MS[\lambda - \delta]$$

$$\text{required } \lambda = \delta(t, s_t)$$

so delta cancel each other

$$f_z = 0$$

$$L_{[t]}^{(1)}(x) = -[(-M\theta)\Delta + 0 \cdot x]$$

$$\text{so } L_{[t]}^{(1)}(x) = M\theta(t, s_t)\Delta$$

$$L^{(1)} \geq M\theta\Delta$$

Second-Order Operation

$$\text{We know } L_{[t]}^{(2)}(x) = -[f_t\Delta + f_z x + \frac{1}{2} f_{zz} x^2]$$

$$\text{From Linear Approx: } f_t = -M\theta, f_z = 0$$

solve f_{zz} :

$$f_z = MS(\lambda - \delta)$$

$$f_{zz} = M \frac{\partial}{\partial S} [S(\lambda - \delta)]$$

$$f_{zz} = M \left[\frac{\partial S}{\partial z} (\lambda - \delta) + S \frac{\partial (\lambda - \delta)}{\partial z} \right]$$

$$\text{because } \frac{\partial S}{\partial z} = S, \text{ and } \lambda \text{ is const. } \frac{\partial \lambda}{\partial z} = 0$$

$$\text{Given } \gamma = \frac{\partial^2 P}{\partial S^2} = \frac{\partial \delta}{\partial S}$$

$$\frac{\partial \delta}{\partial z} = \frac{\partial \delta}{\partial s} \frac{\partial s}{\partial z} = r s.$$

$$f_{zz} = M[s(\lambda - \delta) - s(rs)].$$

$$f_{zz} = M[s(\lambda - \delta) - rs^2]$$

$$r = \delta$$

so.

$$f_{zz} = -Mr s_t^2$$

hence

$$L_{t+1}^{\text{quad}}(x) = -\left[-M\theta\Delta + \theta + \frac{1}{2}(-Mr s_t^2)x^2\right]$$

$$L_{t+2}^{\text{quad}}(x) = M\theta(t, s_t)\delta + \frac{1}{2}Mr(t, s_t)s_t^2 x^2$$